Basic, Scientific & Unit Calculator

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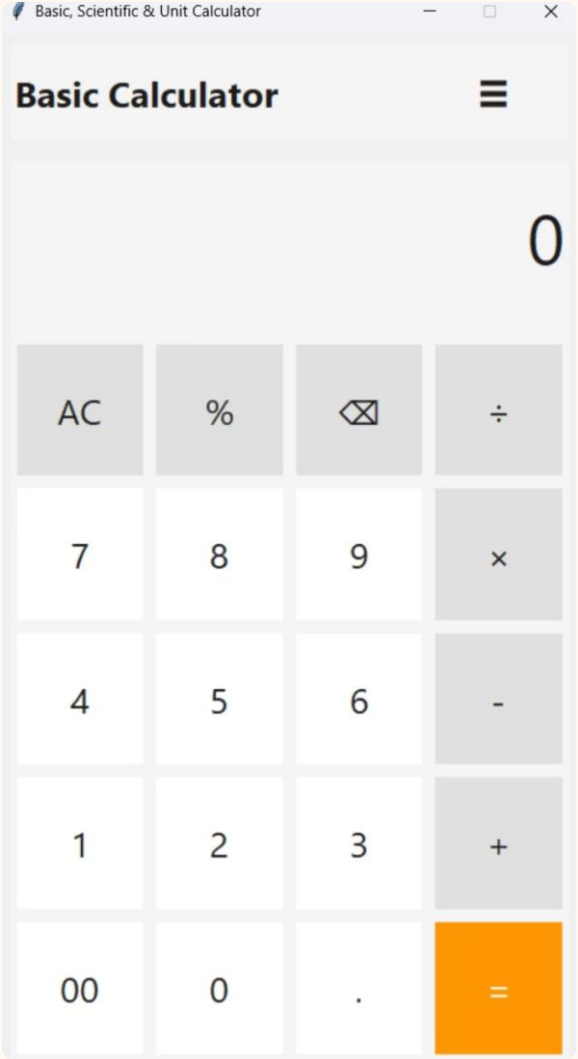
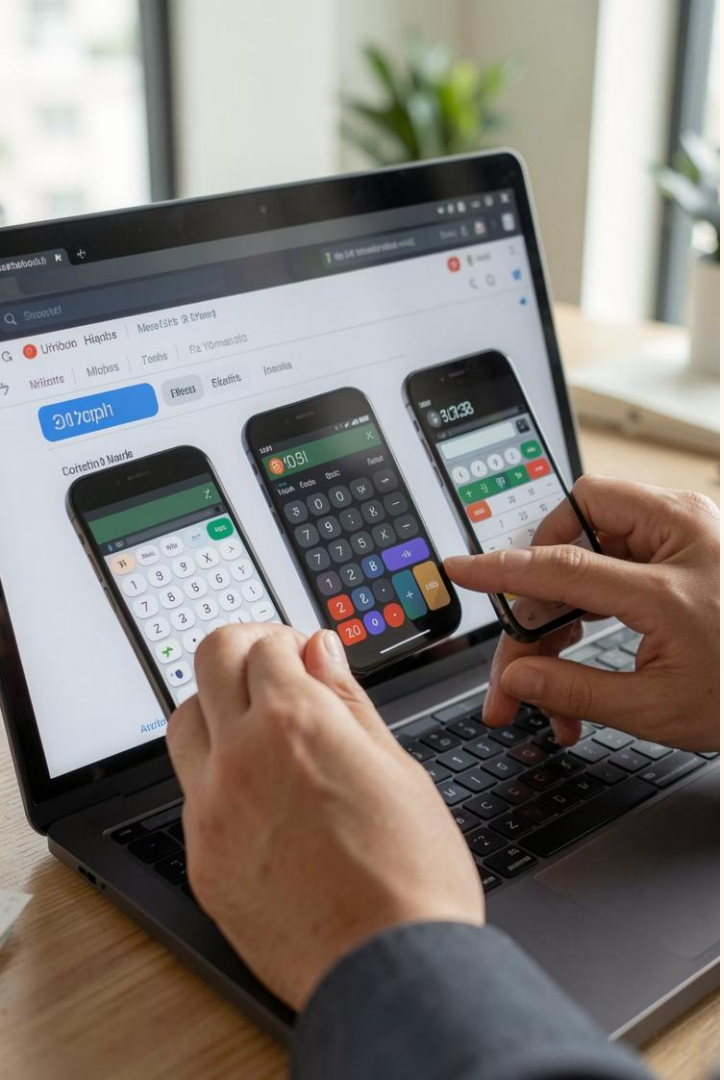
Project Description : A Python Desktop Application for Basic, Scientific, and Unit Calculations.

PYTHON | TKINTER | OOP





Problem Statement

Users often need separate tools for basic calculations, scientific operations, and unit conversions. This increases time, complexity, and reduces productivity.

Time Consumption

Context switching between apps adds friction during study and coding sessions.

Different Interfaces

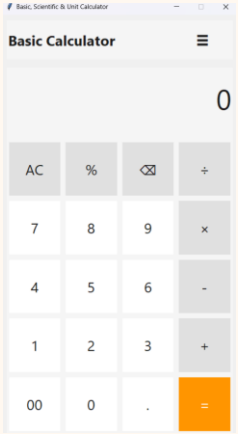
Different controls and layouts cause errors and slow learning.

Multiple Tools Required

No single desktop tool combines basic, scientific, and unit conversion needs in one reliable package.

Project Overview

A Python + Tkinter application that combines Basic Calculator, Scientific Calculator, and Unit Converter modules in one interface.

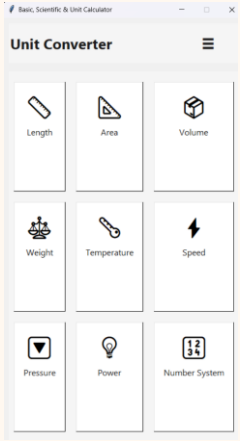


Basic Calculator

Essential arithmetic, percentage, backspace, and a clean numeric keypad for fast entry.

Scientific Calculator

Trigonometry, logarithms, factorials, radians/degrees — designed for engineering coursework and quick prototyping.



Unit Converter

Comprehensive categories: length, area, volume, weight, temperature, speed, pressure, power, and number systems — easy selection and accurate results.

Technology Stack

Python

Core programming language with simple syntax and extensive libraries.

Tkinter

Built-in GUI toolkit for creating desktop applications.

JSON

Stores conversion data and application configurations.

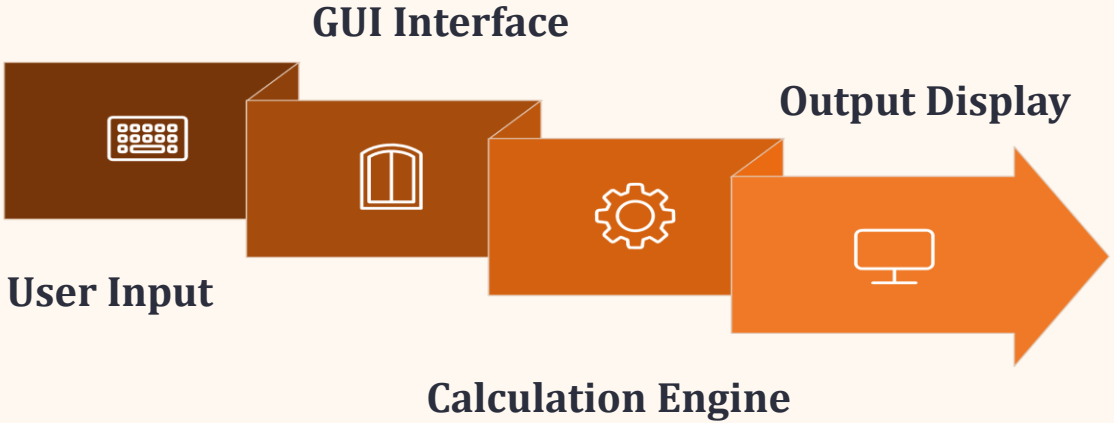
Math Library

Provides scientific functions such as trigonometry, logarithms, and factorials.

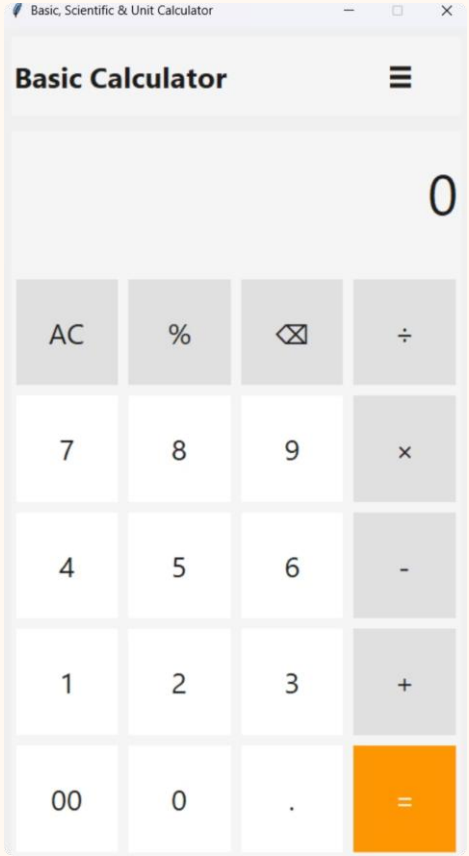
OOP Architecture

Modular classes separate the GUI, calculation engine, and converter logic for better maintainability.

System Architecture



The architecture separates the user interface,
calculation logic, and conversion services,
making the application modular and easy to maintain.



Basic Calculator Features

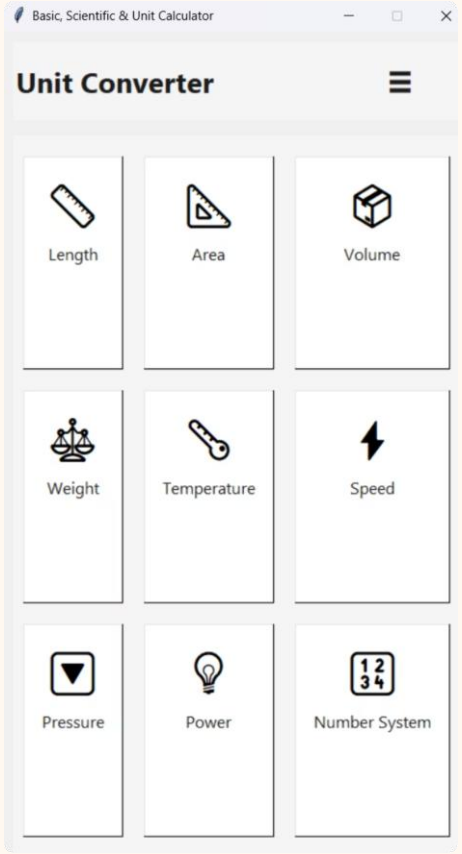
- Core arithmetic: + − × ÷ with a large keypad and keyboard support
- Percentage calculations and double-zero (00) entry for speed
- Backspace / clear (AC) and error-safe input handling
- Simple and user-friendly interface for everyday calculations

Design emphasizes quick numeric entry and minimal visual clutter — ideal for classroom and interview demonstrations.

Scientific Calculator Features

- Trigonometric functions: sin, cos, tan, with rad/deg toggle.
- Logarithmic functions: log (base 10), ln.
- Advanced operations: x^y, factorial, reciprocal (1/x), and π and e constants.
- Floating-point handling and input validation for robust results.

Mode toggle (rad/deg) and a memory-friendly layout support academic test problems and quick formula verification.



Unit Converter Features

Comprehensive, JSON-driven unit tables enable accurate conversions across common categories:

- Length, area, volume
- Weight, temperature
- Speed, pressure, power
- Number system conversions (binary/decimal/hex)

Extensible: add or update units by editing JSON; conversion functions normalize via base units for consistency.

Demo & Results

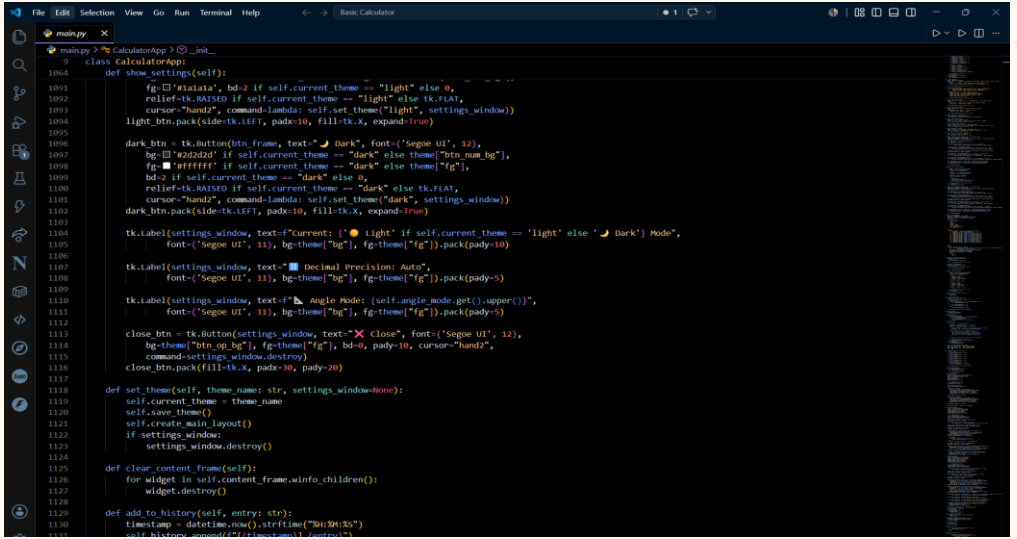
Live demo highlights real-time calculations, accurate scientific operations, and fast unit conversions across all modules.

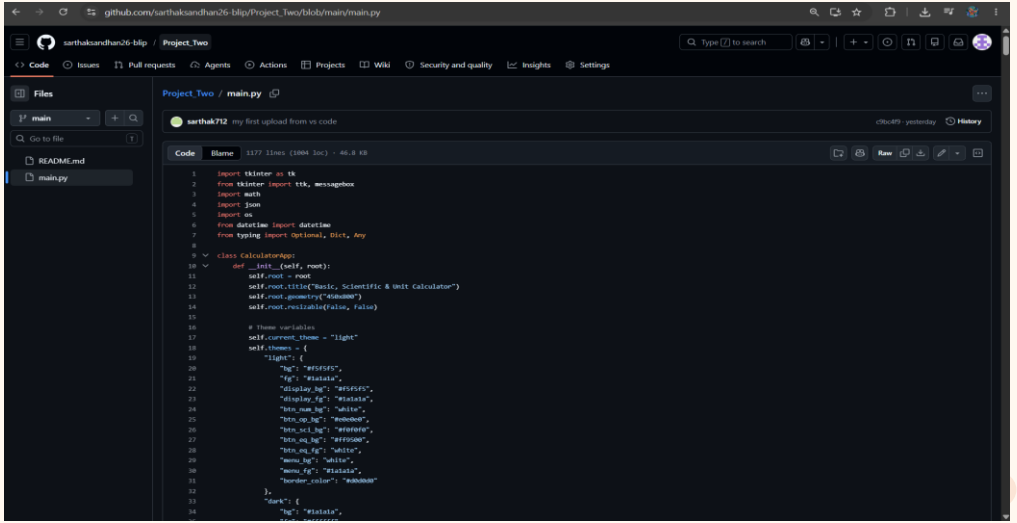
📹 Project Demo Video

Scan to Watch Demo

💻 Project Source Code

Scan to Access Repository





Conclusion & Future Scope

Project achievements: integrated three calculators into a single, maintainable desktop app that demonstrates Python proficiency and thoughtful UX. Skills demonstrated include OOP design, GUI development, and data-driven configuration.

1

1. Improve UI/Accessibility

Keyboard-first navigation, higher-contrast modes, and localization support.

2

2. Add Persistence & Settings

Save user preferences, history, and favorite conversions using JSON or SQLite.

3

3. Packaging & Distribution

Build installers (PyInstaller) for Windows/macOS and add CI to publish releases.

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Thank You
Questions & Discussion